CHENNAI INSTITUTE OF TECHNOLOGY
(Autonomous)

COURSE CODE	**CS4302**
COURSE NAME	**OPERATING SYSTEM**

SLOT 5

1A	List and briefly describe any three operating system services.	5
1B	Compare monolithic and microkernel operating system structures.	5
2A	A task can be divided into 5 equal subtasks and run on a 5-core processor. Each subtask takes 100 ms. Due to overhead (10 ms per core used), total time increases.Calculate the total execution time if: Only 1 core is used All 5 cores are used Also, determine the speedup achieved when using 5 cores vs. 1 core.	5
2B	Explain how virtual machines are created and maintained by OS?	5
3A	Compare various structuring methods for operating systems.	5
3B	Suppose we have two printers connected to a system. For a print job we may allocate any of the two printers. We wish to use semaphores. Describe your design and explain how the scheme shall work. Give a brief sketch of the script that would control printer operation.	5
4A	Utilize the impact of OS evolution on system architecture.	5
4B	Briefly explain the main elements of a computer system and their roles.	5
5A	Outline the basic steps involved in the system boot process.	5
5B	Examine the various design considerations for multiprocessor and multicore systems.	5

SLOT -4

1A	Assume a Round Robin CPU scheduling algorithm with a **time quantum of 5 ms**. There are **100 processes**, each requiring exactly **25 ms** of CPU time. Calculate the **total number of context switches** that will occur before all processes finish. Assume no I/O wait time and that context switch overhead is negligible. Explain the implications of quantum size on system performance...	5
1B	Discuss the following pairs of scheduling criteria conflict in settings. (i) CPU utilization and response time(4) (ii) Average turnaround time and maximum waiting time.(6)I/O device utilization and CPU utilization.(4)	**5**
2A	Consider the 5 Processes, A,B,C,D and E, as shown in the table. The highest number has low priority. Find the completion order of the 5 processes under the policies.	**5**

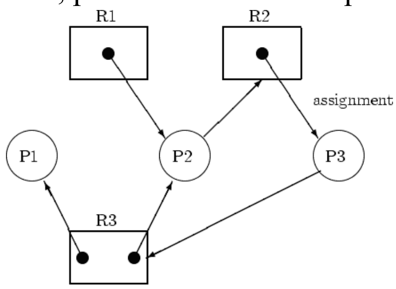
Process	Arrival Time	Burst Time	Priority

A	0	6	3
B	2	4	4
C	4	2	3
D	7	4	2
E	11	2	1

i)Draw four Gantt charts illustrating the execution of these processes using FCFS preemptive SJF, non-preemptive Priority and RR(Quantum=2) scheduling.

2B	Analyze the role of the banker's algorithm in deadlock avoidance.	**5**
3A	Represent and explain the drawback of the typical Semaphore solution to Dining Philosopher's problem with pseudocode and also provide a solution to remedy the drawback.	**5**
3B	Briefly explain the Producer-Consumer problem and its solution using semaphores.	**5**
4A	Explain multiprogramming and the various types of parallelism with neat sketches	**5**

4B Consider the following resource allocation graph :
Determine if there is a deadlock. If so, indicates the processes and resources involved. Show how the deadlock can be resolved through addition of resources. If not, argue why this is the case, i.e there is no deadlock. In either case, provide a feasible sequence of processes to show completion. **5**



5A Consider the following system snapshot using the data structures in the Banker's algorithm, with resources A,B,C, and D, and processes P0 to P4. Answer the following questions using Banker's algorithm **5**

(i) What is the content of the matrix Need?

(ii) Is the system in a safe state?

If a request from a process P1 arrives for (0,4,2,0) can the request granted immediately?

	Max				Alloc				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	?	?	?	?	3	2	1	1
P1	1	7	5	0	1	1	0	1								
P2	2	3	5	6	1	2	5	4								

P3	1	6	5	3	0	6	3	3								
P4	1	6	5	6	0	2	1	2								

5B	In a multicore processor, 4 threads are scheduled on 2 cores. Thread execution times: T1 = 10 ms, T2 = 15 ms, T3 = 20 ms, T4 = 25 ms. Assume two scheduling strategies: (a) Static thread allocation (2 threads per core) (b) Dynamic load balancing with preemption every 5 ms Calculate total completion time under both strategies and determine which strategy gives better performance. C) Describe in detail the design issues and challenges in multicore programming.	5

SLOT 3

1A	Compare and contrast paging and segmentation memory schemes.	5
1B	Free memory holes of sizes 15K, 10K, 5K, 25K, 30K, 40K are available. The processes of size 12K, 2K, 25K, 20K is to be allocated. How processes are placed in first fit, best fit, worst fit. Calculate internal as well as external fragmentation	5
2A	A certain computer provides its user with a virtual memory space of 2^32 bytes. The computer has 2^35 bytes of physical memory. The virtual memory is implemented by paging the page size is 4096 bytes. A user process generates the virtual address 11123456. Explain how the system establishes the corresponding physical location.110. Calculate page faults for (LRU, FIFO, OPT) for following sequences where page frame is three. 0,1,2,1,4,2,3,7,2,1,3,5,1,2,5..	5
2B	Evaluate the various page replacement algorithms with examples.	5
3A	Describe contiguous memory allocation. Discuss its advantages, disadvantages, and challenges.	5
3B	Compare internal and external fragmentation with examples.	5
4A	Justify the role of TLB in improving memory access time.	5
4B	Classify the advantages and challenges of virtual memory systems.	5
5A	Define swapping and explain its role in memory management.	5
5B	Explain the concept of paging and describe the structure of the page table and page replacement algorithms used in virtual memory management.	5

SLOT-2

1A	Explain the concept of file system mounting and its purpose and the importance of disk management and briefly describe how disks are formatted.	5
1B	Select a disk drive has 400 cylinders, numbered 0 to 399.The driver is currently serving a request at cylinder 143 and previous request was at cylinder 125.The queue of pending request in FIFO order is: 86,147,312,91,177,48,309,222,175,130.Starting from the current head position what is the total distance in cylinders that the disk to satisfy all the pending request for each of the following disk scheduling algorithms? a] SSTS b] SCAN c] C-SCAN Categorize the various disk scheduling algorithms with examples.	5
2A	Describe different directory implementation techniques. How do these techniques impact file lookup efficiency in large-scale systems?	5
2B	Given a disk queue with the following I/O requests (in cylinders): **98, 183, 37, 122, 14, 124, 65, 67.**The initial head position is **53.** Calculate the total head movement using **FCFS** scheduling algorithm.	5
3A	Imagine Mr.Z, is working as a Systems Engineer in a company developing a real-time surveillance and storage management system for a smart city. The system captures continuous video streams from thousands of CCTV cameras distributed across various city zones. The architecture must ensure high reliability, efficient disk utilization, fault tolerance, and real-time data access to meet the operational demands of law enforcement and city administration. a) Explain how these physical disk design decisions affect performance, reliability, and data durability for continuous high-throughput video writes. b) Based on the system's requirement for real-time video logging, which algorithm would Guhan recommend and why? c) Implement disk space management strategies such as optimize storage use and reduce fragmentation over time?	5
3B	Imagine you're working in the company W&K TECH LIMI, working on a real-time video conferencing app that supports video calls, file sharing, chat, and screen sharing. These features run at the same time using processes and threads. Recently, the app has become slow—especially video and chat features—when many users are online, and the CPU is heavily used. a) Explain how you would use processes and threads to divide the work in this app. b) Why would you use threads for chat but processes for video encoding? c) How do process operations (like create, suspend, or stop) affect app performance during high load? d) How can inter-process communication (IPC) (like message passing or shared memory) help video and chat work together smoothly?	5
4A	Analyze the importance of free space management in modern file systems. and the role of I/O hardware in managing input and output operations.	5
4B	What are the various file access methods? Briefly explain any two. and significance of file sharing and protection in a multi-user system?	5

5A	Compare sequential and indexed allocation methods with examples. Analyze the design considerations for efficient I/O systems	5
5B	Choose the advantages and limitations of hierarchical file systems. Make use of the role of kernel I/O subsystems in operating systems.	5

SLOT 1

1A	Explain the concept of address space layout randomization (ASLR).	5
1B	In a UNIX system, each file has permissions for owner, group, and others (r/w/x). Calculate the **total number of distinct permission combinations** possible for a file. Assume each of the 3 permission bits (r, w, x) is either 0 or 1 for each category. List at least 3 meaningful combinations. Explain how chmod 755 and 777 relate to these..	5
1C	UNIX file permissions are assigned using a **3-digit octal code** (e.g., 755). Each digit (0–7) represents a combination of **read (4), write (2), and execute (1)** for owner, group, and others. How many total **unique permission combinations** are possible for one file? List how many combinations give **read and execute** but not write for all three categories. Explain real-world use of 755, 700, and 600..	5
2A	Explain the working and advantages of a commercial security kernel.	5
2B	Analyze the impact of mobile operating system (OS) features on user security.	5
3A	Explain the structure of kernel in UNIX operating system?	5
3B	Explain the role of the passwd**,** /shadow**, and** group files in UNIX authorization. What would be the impact if permissions of these files were improperly set? Illustrate with an example..	5
4A	A UNIX server receives **1000 unauthorized access attempts** daily. If each attempt generates a **250-byte log entry,** estimate the **weekly log size** in MB. Assume that 2% of logs are valid alerts, calculate the number of **true positive alerts per week.** Why is it important to filter false positives in logs? Mention tools that help monitor security logs in UNIX. Describe in detail the impact of mobile OS vulnerabilities on user data.	5
4B	Explain the importance of retrofitting security in legacy systems.	5
5A	Classify and explain the different types of Virtual Machines. Compare System VMs and Process VMs with suitable examples. What are the security and usability features of iOS in comparison to Android?	5
5B	Explain the historical evolution of virtual machines and their significance in modern computing. Compare UNIX and Windows security models with examples.	5

SLOT 6

1	A system executes 1000 instructions. 10% are system calls, each taking 50 microseconds. **Calculate the total time spent on system calls.** Assume each normal instruction takes 1 microsecond.	2
2	A monolithic kernel has all services combined and takes 20 MB. In a layered design, the OS splits into 5 layers, each adding 2% overhead. **Calculate the total space used in the layered design.**	2
3	A program performs 200 system calls per second. Each system call takes **1.5 milliseconds** to execute. **Calculate total time spent on system calls per second.**	2
4	A system has 3 modes: **User, Kernel, Monitor**. If a privilege level is denoted as User = 0, Kernel = 1, Monitor = 2, and a process in mode 0 tries to execute an operation requiring level 2, **Is access allowed? Justify with a yes or no and a reason.**	2
5	What is POST and BIOS in the system boot process ?	2
6	Consider following processes with length of CPU burst time in Milliseconds Process Burst time P1 5 P2 10 P3 2 P4 1 All process arrived in order p1, p2, p3, p4 all time zero a) Draw Gantt charts illustrating execution of these processes for SJF and round robin (quantum=1) b) Calculate waiting time for each process for each scheduling algorithm c) Calculate average waiting time for each scheduling algorithm	2
7	Which multithreading model is most efficient and why?	2
8	A system with 4 cores uses a round-robin scheduling algorithm with a time quantum of 5 ms.Given 6 processes with the following burst times: P1 = 12 ms, P2 = 9 ms, P3 = 16 ms, P4 = 7 ms, P5 = 13 ms, P6 = 10 ms. Simulate the scheduling for 2 rounds. Show how the processes are distributed among the cores and compute the remaining burst times.	2
9	Define multicore programming with an example.	2
10	Why is preemptive scheduling preferred in certain scenarios?	2
11	State two key benefits of using segmentation combined with paging.	2
12	What does the term 'thrashing' refer to in memory systems?	2
13	Why is the Translation Lookaside Buffer (TLB) important in paging systems?	2
14	Select two major challenges that arise while managing virtual memory and propose possible solutions.	2
15	Explain how an operating system uses swapping to manage memory under load.	2
16	Given a file stored using contiguous allocation: Starting block = 12, File length = 5 blocks. Find the block numbers occupied by the file..	2
17	In indexed allocation, a file uses a single index block that contains the pointers: 4, 7, 9, 11.How many disk I/O operations are required to read the 3rd block of the file?.	2

18	A file is to be stored using **linked allocation.** The blocks assigned are 15 → 9 → 24 → 18. What block will the file system access after reading block 9?	2
19	A system uses bit vector free space management. If disk blocks 2, 4, 5, 6 are free, show the bit vector for the first 8 blocks.	2
20	Define the term "file protection" in operating systems.	2
21	**What are some common security threats in a Unix environment?**	2
22	What is the role of a hypervisor in virtualization? Differentiate between Type 1 and Type 2 hypervisors.	2
23	Infer the UNIX protection model ensures process and file security using user IDs, group IDs, and permission bits?	2
24	Why is UNIX security still relevant in modern operating systems?	2
25	Relate the concept of retrofitting in the context of security.	2